

HYESAN, Korea, Democratic People's Republic of

WMO: 470160

Lat: 41.3998N

Lon: 128.1796E

Elev: 2343

StdP: 13.49

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-12.8	-21.6	1.8	-14.8	-18.2	2.2	-11.3	16.4	4.2	13.2	7.5	1.0	50	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	86.3	69.1	83.6	67.9	81.0	67.0	72.9	82.1	71.4	79.7	69.9	77.3	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	121.0	77.7	68.7	115.0	76.0	67.3	109.5	74.4	38.2	82.3	36.8	79.7	35.5	77.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	10.1	7.8	DB	-22.6	91.3	2.8	2.8	-24.6	93.4	-26.3	95.0	-27.8	96.6	-29.9	98.7	(4)
(5)			WB	-22.6	76.1	2.8	3.0	-24.6	78.2	-26.3	80.0	-27.8	81.6	-29.9	83.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	3.9	11.8	27.2	43.1	55.5	64.6	71.2	69.6	58.1	44.1	25.3	8.9	(6)
(7)		DBStd	24.42	7.82	9.09	9.10	7.99	6.33	4.85	4.25	5.12	6.00	7.34	10.01	8.66	(7)
(8)		HDD50	5693	1430	1069	708	232	23	0	0	0	8	209	741	1273	(8)
(9)		HDD65	9391	1895	1489	1173	656	298	65	5	18	215	648	1191	1738	(9)
(10)		CDD50	2199	0	0	0	26	194	437	657	607	253	25	0	0	(10)
(11)		CDD65	422	0	0	0	0	4	52	196	160	10	0	0	0	(11)
(12)		CDH74	3196	0	0	0	18	188	567	1342	988	90	3	0	0	(12)
(13)		CDH80	797	0	0	0	0	34	122	396	238	7	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.3	1.7	2.3	2.2	2.2	1.5	1.1	1.1	1.2	1.3	1.2	1.2	(14)
(15)	Precipitation	PrecAvg	25.5	0.3	0.4	0.6	1.3	2.7	3.9	6.5	4.6	2.4	1.1	1.0	0.6	(15)
(16)		PrecMax	33.5	0.8	1.5	1.2	2.7	6.0	6.9	12.9	8.5	5.6	2.2	2.8	1.9	(16)
(17)		PrecMin	16.8	0.0	0.1	0.1	0.4	1.0	1.0	1.9	1.1	0.3	0.1	0.2	0.1	(17)
(18)		PrecStd	4.1	0.2	0.3	0.3	0.6	1.1	1.6	2.5	1.9	1.5	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	40.7	58.4	76.6	84.0	86.8	92.1	89.6	80.7	71.4	53.2	33.6	(19)
MCWB			27.2	34.5	46.1	56.5	60.6	65.2	72.0	71.6	65.1	55.1	43.4	29.5	(20)	
2%		DB	24.5	35.5	51.4	69.6	79.0	83.4	87.3	85.3	76.6	66.0	47.4	29.5	(21)	
		MCWB	21.4	30.6	40.5	51.1	58.2	63.2	70.5	70.8	62.4	51.9	40.3	26.3	(22)	
5%		DB	20.5	31.9	46.6	64.5	74.9	80.4	84.2	82.4	73.4	61.7	43.0	25.8	(23)	
		MCWB	17.9	27.3	37.6	48.5	56.1	63.2	69.7	69.6	60.7	49.2	37.6	23.0	(24)	
10%		DB	16.9	28.2	41.8	59.2	70.8	77.0	81.4	79.6	70.3	57.5	39.0	22.5	(25)	
		MCWB	14.7	24.6	34.9	45.4	54.6	62.0	68.8	68.3	59.1	47.3	34.9	20.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	35.1	46.8	58.0	64.6	69.2	75.4	74.9	68.7	57.1	45.4	30.8	(27)
MCD B			30.0	38.9	56.0	71.1	75.9	79.8	86.1	85.2	75.3	68.3	50.3	33.0	(28)	
2%		WB	21.6	30.6	41.5	52.8	60.8	66.8	73.4	72.9	65.7	53.6	41.1	26.6	(29)	
		MCD B	24.2	34.9	49.9	66.2	72.9	77.4	83.2	81.7	72.8	63.2	46.6	29.3	(30)	
5%		WB	17.9	27.5	38.2	49.4	58.3	65.0	71.8	71.4	63.0	51.0	38.1	23.2	(31)	
		MCD B	20.3	31.3	45.1	61.9	71.8	75.4	80.7	79.4	70.0	59.5	42.3	25.7	(32)	
10%		WB	14.7	24.5	35.1	46.6	56.0	63.5	70.4	70.1	60.9	48.3	34.9	20.2	(33)	
		MCD B	16.8	28.3	41.3	57.9	68.1	73.5	78.5	77.2	67.5	55.8	38.7	22.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	22.3	24.2	21.6	23.6	24.7	22.1	18.3	18.4	22.7	23.2	19.1	20.1	(35)	
5% DB		MCD BR	22.9	25.2	27.3	34.1	34.3	29.5	24.6	22.3	26.2	29.8	24.1	20.6	(36)	
		MCWBR	20.2	20.3	18.0	18.4	16.3	12.9	10.6	10.1	13.6	17.1	17.0	17.7	(37)	
		MCD BR	22.6	23.6	24.8	30.7	29.8	23.7	19.9	18.6	21.2	26.7	21.9	20.0	(38)	
(39)		5% WB	MCWBR	20.0	19.4	17.4	18.1	16.0	11.6	9.9	9.6	13.1	16.4	16.3	17.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.389	0.467	0.541	0.576	0.594	0.531	0.481	0.427	0.428	0.374	0.333	(40)	
taud		2.079	1.932	1.802	1.726	1.710	1.738	1.972	2.101	2.168	2.037	2.072	2.085	(41)		
Ebn at Noon		250	249	242	235	230	226	240	249	254	234	229	235	(42)		
Edh at Noon		35	48	61	71	74	72	56	48	42	43	35	32	(43)		
(44)	All-Sky Solar Radiation	RadAvg	785	1097	1421	1544	1662	1593	1483	1453	1352	1058	721	646	(44)	
(45)		RadStd	43	52	81	95	149	153	155	126	128	92	74	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page